

Business Analytics Fundamental Concept

Terminology Assignment ①

① Business Analytics :- it is method where by Data is used to assist one to make decision in business.

Example : KFC with purchase order number stored in the datawarehouse can tell which product in sales need attention upscale or remove from the shelves.

Source : internal transaction data; international Journal of business Analytics online ISSN : 2334-4558

② Data Analytics : A quantitative data/report by means of semantic modeling and features to build visuals and presentation with insights and storyteller to demonstrate meaning of number.

Example : Bright tv can now see the viewership trend and churn. Develop algorithm and customize ~~at~~ viewership

Source <https://informa.com/service/2026-ai-and-data-analytics>

③ Business Intelligence (BI)

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- This is a co-ordination of variables brought to work together to produce a report with insight to put up a dashboard for continuous reporting

Example: A CEO can see business trends at any angle at his/her command.

Source: Microsoft Learn.

④ Data-Driven Decision making (Triple D)

- quantitative business decision based on numbers

Example: Supermarket very busy, high revenue (sales) but high expenditure. By looking at people coming to buy or steal can understand the business

Source: Harvard Business Review

⑤ Big data - main can be data in warehouse a database unclear not processed (The 3 V's)

Example: People withdrawing money from ATMs

Source - IBM

⑥ Descriptive analysis - reporting in a simple language that even not so tech person can understand

Example: using alias on reporting or customizing your report to local language. You are in South Africa and report to market in China

Source: Microsoft Learn.

3) **Diagnostic Analysis:** explain what happened and what is the root cause and the immediate impacts

Example - Road closure to accident occurred and people arriving late at work.

Source: Wkensy

4) **Predictive Analysis:** - the inherent risk associated with the activity.

Example = when you drive at high the chances are when accident happens the chances are same.

Source: IBM.

5) **Prescriptive Analysis** - to avoid the worst, what should we do.

Example: Re-route the traffic to alternate road.

Source: Wkensy

6) **Data literacy** - it is a skill, ability to read, understand and read chart, question bad data.

Example: Boring report with jargon. Explain FOT cost or DSCR (Debt service Ratio to Managers)

Source: Tableau.

② Data, Measurement & Reporting

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Structured Data - It is a data that stored in rows and columns to make analysis easy

Example - Data in CSV or excel spreadsheets

Source : IBM

⑫ Unstructured Data - Data not in row & column but any other format, make it not understood in SQL language or Python

Example - Data comes in form of pic or video

Source : IBM

⑬ Data warehouse - normally in company we call it server room when system control is at Multi-ple layer of data are stored

Example Onsite is isolated room with air conditioner & keep it cool, limited access per use.

Source : Oracle

⑭ Data Mining - Ability to crack patterns stored in a raw data

Example : Sale trends over period

Source : Investopedia

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15 Data Visualization - converting number/raw data, in charts, graph etc

Example. Pie Chart - show distribution percentage of less than six variable

Source : Tableau

16) Dashboard - It is an executive summary made of one to three pages that tells the story of a business in seconds

Example. Twin stress dashboard for KPI & metric and the second page showing drivers,

Source : Microsoft

17) Metric - Single number show if things are working or broken. (Leading, lagging and healthy metric)

Example. number of truck loading on site. With movement we can analyse our metrics

Source : Investopedia

18) KPI (Key Performance Indicators)

- Our barometer of business - set number or target

Example - we know at a certain set grade of ore content in the rock with decrease if the rock is movable at profit

Source : Harvard Business Review

19) Benchmark. The number one compare himself/herself to know if you are winning average or losing

Example: - year on year comparing

Source: Hubspot

20) Segmentation - some says it is cohorts, we say it is splitting big things in smaller groups that behave differently to enable us to manage

Example: Grouping fleet into same brand and same purchase period

Source: Statista

③ Customer and Sales

21) Conversion Rate: - how much of effort or energy you put in actually turns in cash out

Example: when client payment term is longer it will require you to source more finance assistance to keep the machine running

Source: Google Analytics

22) Customer Acquisition Cost (CAC) - how much one spend to get one paying customer.

Example - Need to buy 10 new trucks to service new customer

Source: Hubspot

23) Customer Lifetime Value - this is how much you will profit from that one customer over the time they stay with you.

Example: When that customer spend R1 million for 2 year contract - The customer value is R24 million.

Source: Investopedia

24) Churn rate - it is the percentage of customers are lost over a period.

Example: When customer get better deal elsewhere they move.

Source: Statista

25) Customer Retention Rate - The percentage of customer I am able to keep.

Example: 20% of my customer since we started operation.

Source: HubSpot

26) Net Promoter Score: the likelihood to you customer to recommend you to other business on scale. 0-10

Example: let customer rate you on your survey form.

Source: Brand Campaigns

27) Average Order Value - how much revenue you get from a transaction or shipment

Example = £2,000 per truck.

Source: Shopify

28) Sales Funnel - stages of journey the customer goes through from first meeting to finalizing contracts and purchases

Example: Customer see catalog, getment invoice, sales release payment final money on the bank.

Source: HubSpot

(4) Marketing & Campaigns

29) Marketing Campaign: promoting a service or product through awareness or activities

Example: Parade Machine at exhibition

Source: HubSpot

30) Post Campaign Analysis - evaluating the impact the marketing campaign ended

Example: Customer engagement metrics

Source: Google Analytics

31) Incremental Revenue - extra revenue you earn from separate campaigns to do

Example: Sales increase after the campaign you receive more orders

Source: Mixpanel

32) Return of Investment (ROI) - additional money we get on top what was initial invested

Example - One invest 120 million and get 125 million
ROI is 5 million

Source: Investopedia

33) Return on Ad spend (ROAS) - return of every money spend on advertising

Example: ROAS earn on every 1 spend on ads
Source: Google Analytics

34) Click - Through Rate (CTR) - percentage expression of people who take next step after seeing what you offer.

Example - 5 clicks on ads

Source: Google Analytics

35) Bounce Rate - percentage of people just after one interaction they leave without going deep into engagement

Example: The customer stop at the entrance of the company, talk to the security personnel and refuse to enter the premises even upon advice to do so by the personnel.

Source: Google Analytics

36) A/B testing - Running two scenarios of ~~for~~ product or service using same data to see which one performs better

- Example: Taking sample from same stock pile and ~~take~~ it to 2 different laboratories

Source: Hubspot.

37) Lead Generation - this value creation and value capture

Example: When a customer sees your products the customer immediately wants offtake contract.

Source: Hubspot

38) Marketing Attribution - figuring out which channel or activity caused the signed offtake contract so that you don't waste money elsewhere to improve DSCR.

Example: Email or LinkedIn or Company visit.

Source: Google Analytics.